B.Sc., CSCS –

Bachelor of Computer Science with Cognitive System

Syllabus

Choice Based Credit System - Learning Outcomes Based Curriculum Framework (CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

B.Sc., CSCS - Bachelor of Computer Science with Cognitive System

Choice Based Credit System Learning Outcomes Based Curriculum Framework (CBCS-LOCF)
(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
I	Language Course – I(LC) –Tamil Other Languages	Tamil - I	6	3	3	25	75	100
	English Language Course – I (ELC)	English - I	6	3	3	25	75	100
	Core Course–I (CC)	Programming in C and C++	5	5	3	25	75	100
	Core Practical-I (CP)	Programming in C and C++ Lab	2	2	3	50	50	100
	Core Course–II (CC)	IT Cognition and Problem Solving	3	3	3	25	75	100
	Core Practical-II (CP)	Introduction to Worksheet Lab	2	2	3	50	50	100
	First Allied Course–I(AC)	Applied Mathematics	4	3	3	25	75	100
	Value Education	Value Education	2	2	3	25	75	100
	Total		**30**	**23**				**800**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
II	Language Course –II (LC) -Tamil/Other Languages	Tamil - II	6	3	3	25	75	100
	English Language Course–II (ELC)	English - II	6	3	3	25	75	100
	Core Course–III (CC)	JAVA Programming	5	5	3	25	75	100
	Core Practical-III (CP)	JAVA Programming Lab	2	2	3	50	50	100
	Core Course–IV (CC)	Information Technology Infrastructure Library - ITIL	2	2	3	25	75	100
	First Allied Course –II (AC)	Statistics	4	3	3	25	75	100
	First Allied Course –III (AC)	Operation Research	3	2	3	25	75	100
	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total		**30**	**22**				**800**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
III	Language Course – III (LC) -Tamil/Other Languages	Tamil - III	6	3	3	25	75	100
	English Language Course - III (ELC)	English - III	6	3	3	25	75	100
	Core Course – V (CC)	Relational Database Management System	5	5	3	25	75	100
	Core Practical - IV (CP)	Relational Database Management System Lab	3	2	3	50	50	100
	Core Course – VI (CC)	Infrastructure Management	3	3	3	25	75	100
	Core Practical – V (CP)	Infrastructure Management Lab	2	2	3	50	50	100
	Second Allied Course – I (AC)	Digital Computer Fundamentals	3	2	3	25	75	100
	Non Major Elective I	A. Fundamentals of Information Technology B. Basics of Office Automation	2	2	3	25	75	100
	Activity Based Course	Health and Wellness	-	1	-	-	-	100
	Total		**30**	**23**				**900**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
IV	Language Course–IV (LC) - Tamil/Other Languages	Tamil - IV	6	3	3	25	75	100
	English Language Course – IV (ELC)	English - IV	6	3	3	25	75	100
	Core Course–VII (CC)	Operating System	6	5	3	25	75	100
	Core Practical-VI (CP)	Operating System Lab	3	2	3	50	50	100
	Second Allied Course-III	Microprocessor & Microcontrollers	3	2	3	25	75	100
	Second Allied Practical - I (CP)	Microprocessor Microcontroller Lab	4	3	3	50	50	100
	Non Major Elective II	A. Working Principles of Internet B. Web Designing	2	2	3	25	75	100
	Total		**30**	**20**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
V	Core Course-VIII(CC)	Data Structure and Algorithms	6	5	3	25	75	100
	Core Course-IX(CC)	Computer Networks	6	5	3	25	75	100
	Core Course-X[CC]	Client Relationship Management	6	5	3	25	75	100
	Core Practical-VII[CP]	Computer Networks Lab	4	3	3	50	50	100
	Major Based Elective - I	A. Mobile Computing B. Computer Organization and Architecture C. Computer Graphics	4	3	3	25	75	100
	Skill Based Elective - I	A. Virtualization and Cloud B. Big Data Analytics	2	2	3	25	75	100
	Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	Internship	Internship (Report of the Study – 80 marks Viva Voce – 20 marks)	-	2	-	20	80	100
	Total		30	27				800

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
VI	Core Course-XI[CC]	Software Engineering and Testing	6	5	3	25	75	100
	Core Course-XII[CC]	Python Programming	6	5	3	25	75	100
	Core Practical - VIII(CP)	Python Programming lab	5	4	3	50	50	100
	Major Based Elective-II	A. Artificial intelligence and Machine Learning B. Process Management C. Data Mining and Warehousing	5	3	3	25	75	100
	Project	Project (Dissertation – 80 marks Viva Voce – 20 marks)	5	3	3	20	80	100
	Skill Based Elective -II	A. Web Technology B. Human Computer Interaction C. Network Security	2	2	3	25	75	100
	Extension Activities	NCC, NSS ,YRC ,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
	Total		30	25				700
	GRAND TOTAL		180	140				4800

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	English Language Course - III (ELC)	English - III	6	3	3	25	75	100
	Core Course – V (CC)	Relational Database Management System	5	5	3	25	75	100
	Core Practical - IV (CP)	Relational Database Management System Lab	3	2	3	50	50	100
	Core Course – VI (CC)	Infrastructure Management	3	3	3	25	75	100
	Core Practical – V (CP)	Infrastructure Management Lab	2	2	3	50	50	100
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						Int.	Ext.	Total
IV	Language Course–IV (LC) - Tamil/Other Languages	Tamil - IV	6	3	3	25	75	100
	English Language Course – IV (ELC)	English - IV	6	3	3	25	75	100
	Core Course–VII (CC)	Operating System	6	5	3	25	75	100
	Core Practical-VI (CP)	Operating System Lab	3	2	3	50	50	100
	Second Allied Course-III	Microprocessor & Microcontrollers	3	2	3	25	75	100
	Second Allied Practical - I (CP)	Microprocessor Microcontroller Lab	4	3	3	50	50	100
	Non Major Elective II	A. Working Principles of Internet B. Web Designing	2	2	3	25	75	100
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	Extension Activities	NCC, NSS ,YRC ,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
	Total		**30**	**25**				**700**
	GRAND TOTAL		**180**	**140**				**4800**

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NON MAJOR ELECTIVE – I		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**III**	**A.** Fundamentals of Information Technology **B.** Basics of Office Automation

NON MAJOR ELECTIVE – II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**IV**	**A.** Working principles of Internet **B.** Web Designing

LIST OF ALLIED COURSES		
S.NO		
1.	**First Allied Course I**	**Mathematics**
2.	**Second Allied Course II**	**Applied Physics**

SKILL BASED ELECTIVE – I		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**V**	**A.** Virtualization and Cloud **B.** Big Data Analytics

SKILL BASED ELECTIVE – II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**VI**	**A.** Web Technology **B.** Human Computer Interaction **C.** Network Security

SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMMES

S.NO	PART	TYPES OF THE COURSE	NO. OF COURSE	NO. OF CREDITS	MARKS
1	I	Language Part–I	4	12	400
2	II	English Part–II	4	12	400
3	III	Core Paper	12	54	1200
4		Core Practical	8	19	800
5		Allied Paper	5	12	500
6		Allied Practical	1	2	100
7		Major Based Elective Course	2	7	200
8		Project	1	3	100
9		Internship	1	2	100
10		Activity Based Course	1	1	100
11	IV	Non-Major Elective	2	4	200
12		Skill Based Elective	2	4	200
13		Value Education	1	2	100
14		Environmental Studies	1	2	100
15		Soft Skill Development	1	2	100
16	V	Gender Studies	1	1	100
17		Extension Activities	1	1	100
TOTAL			**48**	**140**	**4800**

* For those who studied Tamil upto 10th +2 (Regular Stream)
+ Syllabus for other Languages should be on par with Tamil at degree level
Those who studied Tamil upto 10th +2 but opt for other languages in degree level under Part I should study special Tamil in Part IV
**Extension Activities shall be outside the instruction hours.
** Internship is 7 to 10 days.

Note:	Internal Marks	External Marks
1.Theory	25	75
2.Practical	50	50
3. Separate passing minimum is prescribed for Internal and External marks		

FOR THEORY

The passing minimum for CIA shall be 40% out of 25marks [i.e.10 marks].

The passing minimum for University Examinations shall be 40% out of 75 marks [i.e.30 marks]

FOR PRACTICAL

The passing minimum for CIA shall be 40% out of 50marks [i.e.20 marks]

The passing minimum for University Examinations shall be 40% out of 50 marks [i.e.20marks]

PROGRAMME OUTCOMES :

- To recognize the value of lifelong learning and the role of science in society.
- To identify the contemporary issues and demonstrate the scientific knowledge through various processes to arrive at solutions.
- To collect, assess and make use of the information in both theoretical and experimental science.
- To apply advanced theoretical and/or experimental methods, including the use of laboratory techniques, numerical methods and computer simulations in various fields and practical problems.
- To disseminate the science information effectively both in written or by orally before any audience.

PROGRAMME SPECIFIC OUTCOMES:

- ✓ To apply the knowledge of Computer Science to identify, analyze problems and to provide effective solution in the area of Computing.
- ✓ To develop skills to excel in the fields of Computer Science and its enabled services in various fields of Government and Private Sectors.
- ✓ To employ modern computer languages and applications for their successful career, and to create platforms to become an entrepreneur.
- ✓ To instill ethical responsibilities, human and professional values and make their contribution to the society.

FIRST YEAR	CORE COURSE – I	COURSE CODE : 25CCCC1
SEMESTER I	PROGRAMMING IN C AND C++	CREDITS : 5

COURSE OBJECTIVES:

- To know the basic concepts of C Programming, control and looping statements.
- To understand the concepts of Arrays, Functions, Structures, Unions and file management.
- To know the basic concepts of C++ Programming
- To know the OOP concepts
- To understand the Input and Output operation and exception handling techniques.

UNITS	CONTENTS	HOURS
I	Introduction to C – Constants, Variables, Data types – Operator and Expressions - Managing Input and Output operations – Decision Making and Branching – Looping Statements.	20
II	Arrays – Character Arrays and Strings – User defined Functions – Structures and Unions – Pointers – File management in C –Pre-processors.	20
III	Basic concepts of Object –Oriented Programming –Benefits of OOP-Object Oriented Languages-Applications of OOP- Structure of C++ Program-Functions in C++.	20
IV	Classes and Objects- Constructors and Destructors – Operator Overloading and Type Conversions - Inheritance: Extending Classes- Virtual Functions and Polymorphism.	15
V	Managing Console I/O operations- Working with Files-Templates-Exception handling.	15

Text Book:

1. Balagurusamy E .,Programming in ANSI C , Sixth Edition, McGraw-Hill, 2018
2. Herbert Schildt., C++ : The Complete Reference by (4th Edition) , 2017

Reference Book:

1. R.S.Bichkar, Programming with C, University Press, 2012
2. Object oriented Programming with C - E.Balagurusamy, Tata Mc GrawHill publishing company Ltd. New Delhi
3. The C++ Programming Language, Fourth Edition, Bjarne Stroustrup, 2015
4. Programming with C - D.Ravichandran, Tata Mc Graw-Hill publishing company Ltd. New Delhi
5. Object oriented Programming in C++ 4 th Edn.Robert Lafore-Macmilan publishing company Ltd.

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WEB RESOURCES :

https://www.tutorialspoint.com/cprogramming/index.html

https://www.prgramiz.com/c-programming

www.tutorialride.com/cpp/managing-io-console-using-c.html

www.w3schools.in › cplusplus › working-with-filesC++ Working With Files - W3Schools

COURSE OUTCOMES :

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Summarize the basic knowledge to develop C and C++ programs.	K1
C02	Manipulate Looping, arrays and functions in C and C++	K2
C03	Apply and write programs for solving real world problems in C and C++.	K3
C04	Create open ,read ,manipulate ,write and close files in C++	K4
C05	Understand the basic input and output concepts and file management system	K5

FIRST YEAR	CORE PRACTICAL - I	COURSE CODE: 25CCCP1
SEMESTER I	PROGRAMMING IN C AND C++ LAB	CREDITS : 3

COURSE OBJECTIVES :

- To provide the necessary knowledge and practical training for step by step computer program development.
- To present the basic concepts in C programming language and to prepare the students to write modular and readable C Programs.
- Know about how to write basic C++ programming.
- Practice how to create classes.
- Learn how to use control statements.

LIST OF PROGRAMS :

1. Write a C program to check whether a given number is even or odd.
2. Write a C program using a switch case to perform the menu-driven operations.
3. Write a C program to accept n numbers from the user and display them in (Ascending, Descending using For loop)
4. Write a C program to generate the Fibonacci series up to n terms using a user-defined function.
5. Write a menu-driven C program to perform the following string operations.
   b. Find the length of a string
   c. Reverse a string
6. Write a C Program to find the maximum number in an array using pointers.
7. Write a program using constructor to initialize data and destructor to display message when object is destroyed.
8. Create a program with two base classes (Internal, External) and one derived class (Result) using multiple inheritance.
9. Create a program where class Person → Employee → Manager Show multilevel inheritance.
10. Write a program to overload + operator to add two complex numbers using a class.

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	To relate the use of language constructor to solve C and C++ programs	K1
CO2	To develop programs for various concepts in c and C++ language	K2
CO3	To understand and trace the execution the list of programs	K3
CO4	To understand the usage of file handling in C and C++ programming.	K4
CO5	To understand the OOP concepts in C++ programming	K5

FIRST YEAR	CORE COURSE - II	COURSE CODE: 25CCCC2
SEMESTER I	IT COGNITION AND PROBLEM SOLVING	CREDITS : 3

COURSE OBJECTIVES:

- To enable the learners to understand the concepts of cognitive process
- To empower the learners with the skills required for virtual collaboration and culturalsensitivity

UNITS	CONTENTS	HOURS
I	Introduction to Cognition: Meaning, cognitive processes, Development of cognitive psychology.	15
II	Perceptual Processes; Attention: Divided attention, Selective Attention, Visual attention and Auditory attention. Consciousness: Varieties, Subliminal Perception. Visual Perception Perceptual Organizational Processes, Multisensory interaction and Integration: Synthesis, Comparing the senses, Perception and Action.	15
III	Memory- Working Memory: Factors affecting the capacity of working Memory. Long Term Memory: Encoding and Retrieval in Long Term Memory, Autobiographical Memory. Memory Strategies: Practice, Mnemonics using Imagery, Mnemonics using organization. Meta cognition :Meta memory, TOT, Meta comprehension.	15
IV	Problem Solving, Reasoning and Decision Making: VUCA World Problem Solving ,Types of problems, Factors that influence Problem Solving, creativity, Reasoning : Inductive and Deductive Reasoning. Decision Making : Heuristics in decision making , representativeness, availability and Anchoring and adjustment. The framing effect, Overconfidence in decisions, The Hindsight Bias.	15
V	Future Skills: Critical thinking, Adaptive thinking, Cognitive Load Management, Designthinking, Virtual Collaboration and Cultural Sensitivity.	15

TEXT BOOKS:

1. Matlin M.W. (2003) 'Cognition' 5th Edition, Wiley Publication.
2. Riegler, B.R., Reigler, G.L. (2008), Cognitive Psychology – Applying the Science of Mind. 2nd Edition, Pearson Education.

REFERENCE BOOKS:

1. Benjafield J G (2007). 'Cognition' 3rd Edition. Oxford University Press.
2. Goldstein B.E.(2008) 'Cognitive Psychology' 2nd Edition, Wadsworth

WEB RESOURCES:

1. https://sjsu.edu/people/mark.vanselst/courses/p135/s1/Kellogg_c1_fall2013.pdf
2. https://jvapartners.com/problem-solving-and-decision-making-in-a-vuca-environment/
3. https://plato.stanford.edu/entries/critical-thinking/

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO#	CO Statement	K-Level
C01	Outline Cognitive process	K1
C02	Reproduce perceptual process	K2
C03	Identify factors affecting memory	K3
C04	Solve different types of problems	K4
C05	Experiment different skills	K5

FIRST YEAR	CORE PRACTICAL - II	COURSE CODE: 25CCCP2
SEMESTER I	INTRODUCTION TO WORKSHEET LAB	CREDITS : 2

COURSE OBJECTIVE S:

- To perform basic calculations and formatting
- To inculcate the knowledge of Macros
- To create applications using VBA code

LIST OF PROGRAMS:

1. Excel worksheet for Formatting, Math function and Text function
2. Excel worksheet for Graph Function
3. Excel worksheet for VLOOKUP, HLOOKUP and other LOOKUPS
4. Excel worksheet for Pivot
5. Unhide all worksheets at one Go
6. Hide All Worksheets except the Active Sheet
7. Protect and Unprotect All worksheets in a Workbook
8. Save each Worksheets as a separate PDF
9. Change the Letter Case of Selected Cells to Upper Case
10. Sort Data by Single and Multiple Columns
11. Highlight Blank Cells with VBA

COURSE OUTCOMES:

CO#	CO Statement	K-Level
CO1	Remember the fundamentals of MS Excel	K1
CO2	Understand the basics of Functions in Excel	K2
CO3	Analyze and use of LOOKUPS and Formatting	K3
CO4	Write and build applications using VBA Code	K4
CO5	Ability to write Macros and implementation data visualization	K5

E.G.S.PILLAY ARTS & SCIENCE COLLEGE (AUTONOMOUS)
NAGAPATTINAM – 611 002. TAMIL NADU
(Accredited by NAAC With "A" Grade)
(MBA – Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University, Tiruchirappalli – 24)

ALLIED MATHEMATICS

for

B.Sc., Computer Science with Cognitive Systems Programme

First Year	ALLIED COURSE - I			Credit: 4
Semester - I	APPLIED MATHEMATICS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- Apply the basic concepts of Differentiation, Integration and their applications.
- Compute mathematical quantities using Numerical methods.
- Explore fundamental concepts in graph theory.

Units	CONTENT	HOURS
Unit I	**MATRICES:** Matrix – Special types of matrices- Scalar multiplication of a matrix – Equality of matrices – Addition of matrices – Subtraction- Multiplication of matrices – Inverse matrix- Rank of a matrix – A system of m homogeneous linear equations in n unknowns.	18
Unit II	**DIFFERENTIATION:** Maxima and Minima (Problems only) – Points of inflexion.	18
Unit III	**DIFFERENTIAL EQUATIONS:** Linear differential equation with constant coefficients- the operation D and D' – Particular integral – specific methods of finding P.I: x is of the form a).e^{ax} b). cosax or sinax where a is a constant.	18
Unit IV	**NUMERICAL DIFFERENTIATION AND INTEGRATION:** Introduction: Newton's Forward difference formula to compute derivatives – Newton's Backward difference formula to compute derivatives (Simple problems only)- The Trapezoidal rule (excluding truncation error in the trapezoidal formula) – Simpson's rule (simple problems only).	18
Unit V	**GRAPH THEORY:** Introduction – Definition of Graphs – Finite and infinite graphs – Incidence and Degree – Isolated vertex, Pendant vertex, and Null graph. Path and Circuits: Isomorphism – sub graphs – Walks, Paths and circuits – Connected graph, Disconnected Graphs, and components- Euler Graph.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Remember and recall the basic concepts of Applied mathematics	K1
CO2	Illustrate the various notions in the respective streams	K2

CO3	Apply the different terminologies of applied mathematics	K3
CO4	Classify the solutions of mathematical problems using peculiar techniques	K4
CO5	Examine the solutions of a mathematical problem	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	T.K. Manickavasagam Pillai & others, Algebra, Volume II, S.V Publications, 1985 Revised Edition.
2.	S.Narayanan,T.K.ManicavachagamPillai, Calculus, (Vol. III) S. Viswanathan Pvt Limited, 2003.
3.	An introduction to Graph theory by S.Arumugam. and S.Ramachandran
REFERENCE BOOKS	
1.	T.K. Manickavasagam Pillai & others, Algebra, Volume III, S.V Publications, 1985 Revised Edition
WEB RESOURCES	
1.	https://www.britannica.com/topic/operations-research
2.	https://byjus.com/maths/linear-programming/
3.	https://www.gatexplore.com/transportation-problem-study-notes/
4.	https://youtu.be/rowWM-MijXU5.https://youtu.be/TQvxWaQnrqI

FIRST YEAR	CORE COURSE - III	COURSE CODE: 25CCCC3
SEMESTER II	JAVA PROGRAMMING	CREDITS : 5

COURSE OBJECTIVES:

- To learn how to implement object-oriented designs with Java.
- To identify Java language components and how they work together in applications.
- To design and program stand-alone Java applications.
- To learn how to use exception handling in Java applications.
- To learn about AWT concepts and Applets.

UNITS	CONTENTS	HOURS
I	Java Language Fundamentals : The Building Blocks of Java – Data Types – Variable Declarations – Wrapper Classes – Operations and Assignment – Control Structures – Arrays	**15**
II	Java as an OOP Language : Defining Classes – Modifiers – Packages – Interfaces-– Strings – String Buffer Class.	**15**
III	Exception Handling : Introduction – Basics of Exception Handling – Exception Hierarchy – Constructors and Methods in Throwable Class - Unchecked and Checked Exceptions – Handling Exceptions in Java – Exception and Inheritance – Advantages of Exception Handling Mechanism.	**15**
IV	Multithreading : Introduction – Creating Threads – Thread Life-cycle Files and I/O Streams : Overview – Java I/O – File Streams – File Input Stream and File Output Stream – File Streams – Random Access File – Serialization.	**15**
V	Applets: Basis - Lifecycle - Applet classes - Application – Graphics; AWT-I: GUI Programming - AWT classes - Windows fundamentals- Creating Windows - Dialog Boxes - Layout Managers - Radio Buttons and Check Boxes – Borders – Java Swing.	**15**

TEXT BOOKS:

1.Object Oriented Programming through Java, P. Radha Krishna, University Press, 2019.

2.S.Sagayaraj, R.Denis, P.Karthik & D.Gajalakshmi, "Java Programming", Universities Press, 2021.

3.Patrick Naughton and Herbert Schildt. "The Complete Reference JAVA 2". 3 rd Edition. Tata McGraw-Hill Edition, 2011.

4. Muthu C. "Programming with JAVA". 2nd Edition. Vijay Nicole Imprints, 2020.

REFERENCE BOOKS:

1. Java Programming, K. Rajkumar, Pearson India, 2013
2. Muthu C. "Programming with JAVA". 2nd Edition. Vijay Nicole Imprints, 2011.
3. Ken Arnold Gosling and Davis Holmen. "The Java Programming Language". 3rd Edition. Addition Wesley Publication.

WEB RESOURCES:

https://www.tutorialspoint.com/cprogramming/index.html

https://www.w3schools.in/dataStructures/intro.html

https://www.programiz.com/java-programming

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO#	CO Statement	K-Level
C01	Remember the fundamentals of programming such as variables, Conditional statements and iterative execution statements.	K1
C02	Understanding the OOPs concepts and String	K2
C03	Apply the concepts of Exception Handling.	K3
C04	Implement the concepts of Multithreading and File Streams.	K4
C05	Analyze the concepts of applet programming, graphics programming, Java swing.	K5

FIRST YEAR	CORE PRACTICAL - III	COURSE CODE: 25CCCP3
SEMESTER II	JAVA PROGRAMMING LAB	CREDITS : 3

COURSE OBJECTIVE S:

- To Impart Practical Training in Java Programming Language.
- To gain practical knowledge in Exception Handling.
- To gain practical expertise in coding Core Java programs.
- To become proficient in the use of AWT.
- To develop a program based on Event Handling and Swing.

LIST OF PROGRAMS:

1. Write a program to sort the given numbers using arrays.
2. Write a program to implement the FIND and REPLACE operations in the given multiple text.
3. Write a program to find the area of a rectangle using constructor.
4. Write a program to find the student's percentage and grade using command line arguments.
5. Write a program to draw circle or triangle or square using polymorphism and inheritance.
6. Implementing User defined Exception Handling.
7. Implement multiple inheritance concepts in java using interface, you can choose your own example of a company or education institution or a general concept which requires the use of interface to solve a particular problems.
8. Write a program to create threads and assign priorities to them.
9. Write a program to develop an applet to play multiple audio clips using multithreading.
10. Write a program to implementing a Calculator using Java Swing.

COURSE OUTCOMES:

CO#	CO Statement	K-Level
CO1	Remember the fundamentals of Java programming language	K1
CO2	Understand the basics of Java programming, multi-threaded programs and Exception handling	K2
CO3	Analyze and use Java in a variety of applications.	K3
CO4	Write and debug a software application developed using the Java Programming language.	K4
CO5	Ability to design application programs using Java	K5

SUB CODE	TITLE	CATEGORY	CREDIT
	INFORMATION TECHNOLOGY INFRASTRUCTURE LIBRARY(ITIL)	**CORE COURSE-IV**	**2**

Course Objectives

- To be able to design a knowledge based system.
- To understand Service lifecycle model
- To know the Key Principles Models and Concepts of service management
- To understand the process management and risk management
- To know the Challenges in providing IT Infrastructure Services
- To understand the event management concepts.

Course Outcomes (COs)

1. Design a knowledge based system.
2. Understand service lifecycle model
3. Know the key principles models and concepts of service management
4. Understand the process management and risk management
5. Know the challenges in providing IT infrastructure services
6. Understand the event management concepts.

Unit I

Introduction ITIL **-** Service Life Cycle Model - What is ITIL - Components and Phases of a Service Life Cycle - Main concept of Service life cycle - Service management as a Practice - IT today and IT opportunity

Unit II

What is a Service - What is Service management - what comprises value - the 4 Ps of Service Design - Key It service management roles **-** Key Principles Models and Concepts

Unit III

Process - Functions - Specific Roles - RACI - Risk Management - Business Case - Life Cycle Phases - Service Strategy - Service Design - Service Transition - Service operation - CSI

Unit IV

Automation - Evolution of IT Infrastructure Services - Challenges in providing IT Infrastructure Services - The future state of IT Infrastructure Services - Automation and Analytics - the approach of the future

Unit V

SNOC - Event Management - Objectives of Event Management - Scope of event management - Value to the organization - Draw IT Infrastructure facilities for Hospital Management System, e
- governance and Banking Sector

Suggested Readings

1. Service Support (CCTA): Part 15 (IT Infrastructure Library)

First Year	ALLIED COURSE - II			Credit: 3
Semester - II	STATISTICS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- Enable the short historical development of Statistics.
- Provide the knowledge to interpret and solve the statistical problems.
- Explore the ideas of statistical tools

Units	CONTENT	HOURS
Unit I	**Importance, Functions, Limitations:** Importance - Statistics in Mathematics - Functions of Statistics–Limitations of Statistics. **Diagrammatic Representation:** Introduction– Advantages–Limitations of a Diagram – Rules for Making a Diagram – Types of Diagram	12
Unit II	**Measures of Central Tendency** Averages –Arithmetic Mean – Median – Mode – Geometric Mean	12
Unit III	**Dispersion:** Dispersion–Measures of Dispersion– Coefficients of Dispersion(Simple Problems Only)	12
Unit IV	**Correlation:** Introduction–Meaning of Correlation– Scatter Diagram–Karl Pearson's Co-efficient of Correlation – Rank Correlation (Derivations not needed and Simple Problems Only). **Linear Regression:** Introduction–LinearRegression– Regression Coefficients–Properties of Regression Coefficients(Derivations not needed and Simple Problems Only)	12
Unit V	**Testing the Hypothesis:** Applications of Chi– Square Distribution –Goodness of Fittest– Applications oft-distribution– t-test for Single Mean – t-test for difference of Means – Applications of F-distribution – F-test for Equality of two Population Variances (Simple Problems Only)	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Remember and recall the basic concepts of statistics.	K1
CO2	Illustrate the various notions in the respective stream.	K2
CO3	Apply the different terminologies of statistics.	K3
CO4	Classify the solution of statistical methods using various techniques.	K4
CO5	Explain the solution of statistical problems.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Pillai.R.S.N & Bhagavathi (2008).*Statistics Theory and Practice*. S.Chand & Sons, New Delhi.

2.	Gupta.S.C. &V.K.Kapoor. (2014). *Fundamentals of Mathematical Statistics*. Sultan Chand &Sons, New Delhi.
REFERENCE BOOKS	
1.	Gupta. S.C. &Kapoor. V.K.(2004). *Elements of Mathematical Statistics*. Sultan Chand & Sons, New Delhi.
2.	Veerarajan.T.(2010). *Probability, Statistics and Random Processes.* Tata Mc Graw Education Private.
WEB RESOURCES	
1.	https://www.youtube.com/watch?v=6DYtC7lrVuY
2.	https://www.youtube.com/watch?v=YGObRCEZiC8
3.	https://www.youtube.com/watch?v=xZ_z8KWkhXE
4.	https://www.youtube.com/watch?v=nk2CQITm_eo

First Year	ALLIED COURSE - III			Credit: 2
Semester - II	OPERATIONS RESEARCH			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- To train the students to solve assignment problems, transportation problems.
- To train the students in network problems.
- To make students solve real life problems in Business and Management.

Units	CONTENT	HOURS
Unit I	**Operations Research:** Introduction - Basics of OR – OR & decision making – Role of Computers in OR - Linear programming formulations & graphical solution of two variables.	12
Unit II	Canonical & standard forms of LPP - Simplex Method : Simplex Method for < , = , > constraints – Charne's method of penalties.	12
Unit III	**Transportation problem:** Transportation algorithm –Degeneracy algorithm – Degeneracy in Transportation Problem , Unbalanced transportation problem- Assignment algorithm.	12
Unit IV	**Sequencing Problem:** Processing of n jobs through two machines – Processing of n jobs through 3 machines – processing of two jobs through m machines.	12
Unit V	**Networks:** Network – Fulkerson's rule - measure of activity – PERT computation – CPM computation.	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Recognize and relate LPP and solving LPP using graphical method.	K1
CO2	Compute Simplex Algorithm and Big-M Method of LPP.	K2
CO3	Explain Transportation problem and Evaluate its initial basic feasible solution and solve assignment problem using Hungarian algorithm.	K3
CO4	Sequencing the jobs to be carried out based on Cost Optimization.	K4
CO5	Describe and Construct Network and compute PERT and CPM.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Manmohan & Gupta, Operations Research, Sultan Chand Publishers, New Delhi.
REFERENCE BOOKS	
1.	Prem Kumar Gupta and D.S. Hira, Operations Research : An Introduction, S.Chand and Co., Ltd. New Delhi.
2.	Hamdy A. Taha, Operations Research (7th Edn.), McMillan Publishing Company, New Delhi, 1982.

WEB RESOURCES	
1.	https://www.britannica.com/topic/operations-research
2.	http://ebooks.lpude.in.operationsresearch

SECOND YEAR	CORE COURSE – V	COURES CODE: 25CCCC5
SEMESTER -III	RELATIONAL DATABASE MANAGEMENT SYSTEM	CREDIT:5

COURSE OBJECTIVE:

To learn the basic concepts of relational database management system, normalization and entity-relationship model.

UNIT	CONTENTS	HOURS
I	Database Management: Introduction - Database System - Database - Data Independence - Relational Systems. Database System Architecture: Introduction - Three Levels of the Architecture - Database Management System - Client/Server Architecture.	12
II	An Introduction to Relational Databases: Introduction - Relational Model - Relations. An Introduction to SQL: Introduction - Views - Transactions - Embedded SQL - Dynamic SQL.	12
III	The Relational Model: Relations - Tuples - Relation Types - Relation Values - Relation Variables. Relational Algebra: Original Algebra - Syntax - Semantics - Examples.	12
IV	Database Design: Functional Dependencies - Introduction - Basic Definitions - Trivial and nontrivial dependencies. Normalization: First, Second and Third Normal Forms - BCNF.	12
V	Higher Normal Forms: Multi-values dependencies and Fourth Normal Form - Join Dependencies and Fifth Normal Form. Semantic Modeling: The E/R Model - E/R Diagrams - Database Design with the E/R Model.	12

COURSE OUTCOMES:

On successful completion of the course the students would gain knowledge about the relational database management system.

Text Book:

1. Silberschatz, A., Korth, H.F., & Sudarshan, S., Database System Concepts, 7th Edition, McGraw Hill Education, 2020.
2. Elmasri, R. & Navathe, S.B., Fundamentals of Database Systems, 7th Edition, Pearson Education, 2017.

Reference Books:

1. Sadalage, P.J. & Fowler, M., *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012.
2. Celko, J., *SQL for Smarties: Advanced SQL Programming*, 5th Edition, Morgan Kaufmann, 2014.
3. Hernandez, M.J., *Database Design for Mere Mortals*, 4th Edition, Addison-Wesley, 2020

Web Sites :

1. **https://www.youtube.com/@nesoacademy**
2. **https://www.youtube.com/@GateSmashers**
3. **https://www.geeksforgeeks.org/dbms/**

SECOND YEAR	CORE PRACTICAL – IV	COURES CODE: 25CCCP4
SEMESTER - III	RELATIONAL DATABASE MANAGEMENT SYSTEM LAB	CREDIT:2

COURSE OBJECTIVE:

To enable the students to learn the concepts of database designing, to present SQL and procedural interfaces to SQL comprehensively and to give a good formal foundation on the relational model of data.

LIST OF PRACTICAL EXERCISES:

EXERCISE	CONTENTS
1.	Query a database using SQL DML and DDL commands.
2.	Perform queries using ANY, ALL, IN, INTERSECT, UNION.
3.	Perform query operations using aggregate functions.
4.	Create views, perform update and delete operations on views.
5.	Create a table and perform trigger operations.
6.	Write a PL/SQL program to find the largest of three numbers.
7.	Write a PL/SQL program to find the factorial of a given number.
8.	Write a PL/SQL program to find the given number is Prime or not.
9.	Write a PL/SQL program to accept the text and reverse the text and check whether the given text is Palindrome or not.
10.	Write a PL/SQL program to find the reverse of a given number

COURSE OUTCOMES:

On successful completion of the lab, the students can be able to design and implement a database schema for a given problem-domain, normalize a database, populate and query a database using SQL DML/DDL commands, and PL/SQL programming.

Web Sites

1. https://www.youtube.com/@nesoacademy
2. https://www.youtube.com/@GateSmashers
3. https://www.geeksforgeeks.org/dbms/
4. https://www.w3schools.com/sql/
5. https://www.tutorialspoint.com/sql/index.htm

SECOND YEAR	CORE COURSE - VI	COURES CODE: 25CCCC6
SEMESTER - III	INFRASTRUCTURE MANAGEMENT	CREDIT:3

COURSE OBJECTIVE:

To understand various concepts of infrastructure planning and management.

UNIT	CONTENTS	HOURS
I	**Introduction to IT infrastructure:** Data Networks and API : Fundamentals, Corporate network Functions, Business uses, Wireless and Mobile Infrastructure, Messaging and Collaboration Technology, IT Risk Management, Financial Crimes and Fraud Defenses, Sustainability and the Triple Bottom Line Approach. **Internet Technologies and Search Strategies:** Search Technology, Organic Search and Search Engine Optimization, paid Search Strategies and Metrics.	5
II	**Information Technology in Business:** Digital Technology Transforming Business Processes, Competitive Advantage and SWOT Analysis. **Building Business Capabilities with Data Governance and Cloud Services:** Data Governance Strategy, Enterprise IT Architecture, Information and Decision Support Systems, Data Centers and Cloud Computing, Cloud services Delivery Model.	5
III	**Data management, Big Data, and Business Intelligence:** Data Management Technologies and Techniques, Transaction Processing and Analytics Processing Systems, Data Ownership and Organizational Politics, data Life Cycle, Big Data Analytics, Data and Text Mining, Data Visualization, Enterprise data Mashups, Digital Dashboards.	5

IV	**Networked Economy:** Impact of Computer Networks on Business, Elements of Networked Economy, using IS Functions to Deal with Business Risks, Privacy, Health and Ethical Issues in Networked Economy, Future of the Networked Economy.	5
V	**Strategies for Successful Infrastructure Project Implementation**: Risk Management Framework for Infrastructure Projects, Shaping the Planning Phase of Infrastructure Projects to mitigate risks, Introduction to Fair Process and Negotiation, Sustainable Development of Infrastructure, Information Technology and Systems for Successful Infrastructure Management, Infrastructure Modeling and Life Cycle Analysis Techniques.	4

COURSE OUTCOMES :

After the completion of this course the students would be able to

1. Design integrated framework for infrastructure planning and management.
2. Analyse the strategies for Infrastructure Project implementation.
3. Perform Infrastructure modelling and Life Cycle Analysis Techniques.

Text Books

1. James A. O'Brien & George M. Marakas, Management Information Systems, 10th Edition, McGraw Hill Education, 2011.
2. Ralph Stair & George Reynolds, Principles of Information Systems, 13th Edition, Cengage Learning, 2018.
3. Paul Bocij, Andrew Greasley & Simon Hickie, Business Information Systems, 6th Edition, Pearson Education, 2015.

Reference Books

1. IT Infrastructure Library (ITIL), IT Service Management Foundation, Axelos Publications, 2019.
2. Thomas Erl, Cloud Computing: Concepts, Technology & Architecture, Pearson Education, 2014.
3. George Spafford & Gene Kim, The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win, IT Revolution Press, 2018.
4. Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, 5th Edition, Pearson Education, 2013.

Web Sites

1. https://www.coursera.org/learn/it-infrastructure-services
2. https://www.geeksforgeeks.org/what-is-it-infrastructure/
3. https://www.youtube.com/@SimplilearnOfficial

SECOND YEAR	CORE COURSE- V	COURES CODE: 25CCCP5
SEMESTER -III	INFRASTRUCTURE MANAGEMENT LAB	CREDIT:3

COURSE OBJECTIVE:

To provide students with practical knowledge in installing, configuring, and managing client-server systems and network services. The course aims to develop skills in user access control, operating system administration, domain management, security policies, backup management, and Active Directory Services in a network environment.

EXERCISE	CONTENTS
1.	**A. Client Server Installation**
	1. Review Hardware Component Installation
	2. Review Server Installation
2.	**B. Access Control and User permission settings**
	1. To set user access control
	2. To create permission setting
	3. To create profile setting etc.
	4. To setup child domain creation
	5. To setup trust relation
	6. To set up group policy setting
	7. To create backups
3.	**C . Operating System Administration**
	1. Installing and Configuring DNS Services
	2. Installing and Configuring DHCP Server Services
	3. Installing and Configuring Remote Access Services
	4. Configuring NAT, ICS
	5. Configuring VPN
	6. Implementing and Monitoring IPSec
	7. Installing and Setting up ADS, Delegation of Control
	8. Study of AGDLP Process

	9. Creating Forest, Domain and Tree (ADS Logical Element)
	10. Creating Site and Site Link (ADS Physical Element)
	11. Study of ADS Commands (dsadd, dsmod, dsmove, dsrm, movetree)
	12. Configuring Group Policy, Assigning and Publishing Objects using GPO.

COURSE OUTCOMES:

On successful completion of this lab, students will be able to

- Install and configure client-server hardware and software components.
- Configure user accounts, access control, permissions, profiles, and group policies in a network environment.
- Create and manage child domains, trust relationships, forests, domains, trees, sites, and site links using Active Directory Services.
- Install and administer network services such as DNS, DHCP, Remote Access Services, NAT, ICS, VPN, and IPSec.

Web Sites:

1. https://www.youtube.com/@GateSmashers
2. https://www.youtube.com/@NesoAcademy
3. https://www.geeksforgeeks.org/computer-network-tutorials/
4. https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm

SECOND YEAR	ALLIED COURSE-I	CREDIT :
SEMESTER - III	ALLIED PHYSICS-I (DIGITAL COMPUTER FUNDAMENTALS)	Subject code:

ALLIED PHYSICS-I
DIGITAL COMPUTER FUNDAMENTALS

Allied physics-I (only for AIML & CSCS students)

Objective :

To Understand the Number system, gates, Electronic circuits and Understand the Electronic combinational systems.

UNIT- I: Number system **(12 Hours)**

Number system – Binary, Octal, Decimal, Hexa decimal – Conversion – Binary codes – BCD code Gray – Binary Addition, Subtraction, Multiplication and Division.

UNIT- II: BOOLEAN ALGEBRA **(12 Hours)**

Boolean Algebra – Boolean Postulates and laws – De-Morgan's theorem – Simplication of Boolean expression – Karnaugh map.

UNIT- III: LOGIC GATES **(12 Hours)**

Logic gates – AND, OR, NOT, NAND, NOR and Ex-OR gates realization of gates using universal gates.

UNIT-IV: COMBINATIONAL CIRCUITS **(12 Hours)**

Arithmetic Circuit – BCD Addition, Subtraction, 1s complement and 2s complement – Half Adder and Full Adder – Half Subtractor and Full Subtractor.

UNIT – V: MEMORY DEVICES **(12 Hours)**

Classification of Memories –Primary Memory: RAM – ROM – Secondary Memory: hard Disk – Floppy Disk – Compact Disk – Digital Versatile Disk(DVD) – Flash Drive.

Total Hours- 60

Book for Study:

Digital Electronics-Meena. K

Text Books

1. M. Morris Mano & Michael D. Ciletti, *Digital Design*, 6th Edition, Pearson Education, 2018.
2. Thomas L. Floyd, *Digital Fundamentals*, 11th Edition, Pearson Education, 2015.
3. Donald P. Leach, Albert Paul Malvino & Goutam Saha, *Digital Principles and Applications*, 8th Edition, McGraw Hill Education, 2015.

Reference Books

1. Ronald J. Tocci, Neal S. Widmer & Gregory L. Moss, *Digital Systems: Principles and Applications*, 12th Edition, Pearson Education, 2017.
2. Charles H. Roth Jr. & Larry L. Kinney, *Fundamentals of Logic Design*, 7th Edition, Cengage Learning, 2014.
3. A.Anand Kumar, *Fundamentals of Digital Circuits*, 4th Edition, PHI Learning, 2016.
4. R.P. Jain, *Modern Digital Electronics*, 5th Edition, McGraw Hill Education, 2018.

Web Sites

1. https://www.youtube.com/@nesoacademy
2. https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/
3. https://www.tutorialspoint.com/digital_circuits/index.htm

22/5/26

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R. Priya 22.5.26

Dr. R. PRIYA
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SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25ITNM1A
SEMESTER - III	FUNDAMENTALS OF INFORMATION TECHNOLOGY	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize the students with the world of IT and IT-enabled services.
- To provide an in-depth knowledge about internet and internet tools.
- To enable the students to understand about Computer Security.

UNIT	CONTENTS	HOURS
I	Introduction to Computers - Generation of Computers - Classification of Digital Computer - Anatomy of Digital Computer.	5
II	CPU and Memory - Secondary Story Devices - Input Devices - Output Devices.	5
III	Introduction to Computer Software - Programming Language - Operating Systems - Introduction to Database Management System.	5
IV	Computer Networks - WWW and Internet - Email - Web Design	5
V	Computers at Home, Education, Entertainment, Science, Medicine and Engineering - Introduction to Computer Security - Computer Viruses, Bombs, Worms.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand basic concepts and terminologies in IT and IT-enabled services.	K2
CO2	Understanding personal computers and their operations.	K2
CO3	Understand about operating systems and database management	K2
CO4	Comprehend about WWW, internet, email and web design concepts	K4
CO5	Respond to computer security issues.	K3

REFERENCES:

1. Alexis Leon and Mathews Leon, Fundamentals of Information Technology, Vikas Publishing House Pvt. Ltd, 2009
2. Fundamentals of Computers and Information Technology, M.N Doja,2005
3. Ramesh Bangia,"Computer Fundamentals and Information Technology", Laxmi Publications Pvt Limited, 2008.
4. Bharihoke, "Fundamentals of Information Technology", Excel Books, 2009. Ralph Stair, George Reynolds,
5. "Fundamentals of Information Systems" Cengage Learning, 2015.

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25ITNM1B
SEMESTER - III	BASICS OF AUTOMATION	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize students with database concepts and data handling using MS-Access.
- To develop practical skills in using MS-Office applications for documentation & database management.
- To provide knowledge of spreadsheet operations, formulas, charts, and data analysis using MS-Excel.

UNIT	CONTENTS	HOURS
I	Introduction to MS – Windows, Operating System, Windows Desktop Icons and their functions, Dialog Boxes, Task Bar, Parts of Windows, Keyboard short keys or hot keys. Introduction to MS-Office.	5
II	MS-Word – creating, editing saving and printing text documents, Font formatting, inserting tables, smart arts, page breaks, working with images, Spelling and Grammar Check, document properties, Mail Merge.	5
III	MS-Excel – spreadsheet basics, creating, editing saving and printing spreadsheets, working with functions & formulas, modifying worksheets with color & auto formats, graphically representation : charts & graphs, data forms, analyzing data – securing spreadsheets	5
IV	MS-Power Point – creating, editing saving and printing slides, auto layouts, custom animation, slide transition, Graphical representation – charts & graphs, slide presentation.	5
V	MS-Access – planning a Database, access screen, creating database, tables, working with forms, creating queries, types of reports, printing & print preview, importing data.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the basic concepts of operating systems, Windows environment, and MS-Office tools.	K2
CO2	Create, edit, format, and print professional documents using MS-Word.	K3
CO3	Prepare spreadsheets, apply formulas and functions, analyze data, and generate charts using MS-Excel.	K3
CO4	Create and manage simple databases, forms, queries, and reports using MS-Access.	K3
CO5	Apply office automation tools effectively for academic, personal, and professional purposes.	K3

REFERENCES:

1. Computer Fundamental & MS Office — Shashwat Publication, 2025 Edition. Covers Windows, MS Word, Excel, PowerPoint and Access aligned with NEP-2020 syllabus.
2. Technology For Success & The Shelly Cashman Series Microsoft 365 & Office Introductory — Cengage, 2025 Edition. Updated for Microsoft 365 and Office 2021 applications.
3. Keyboard 3: Windows 10 and MS Office 2016 — Oxford University Press India, 2025 Edition.

	COURSETITLE	YEAR	SEMESTER	CREDIT
	A.FOOD AND NUTRITIONAL BIOTECHNOLOGY	II	III	2

OBJECTIVES

- The students will be able to learn about food Preservation and microorganisms associated with itLearn about packaging of different foods. Learn types of Foods and their nutritional values
- Learn about components of foods and their deficiency disorders.5. Learn about Essential minerals , BMR and RDA

Unit I

Introduction to Food and Nutrition Biotechnology, Application of biotechnology to improve food quality, safety, production, and Nutritional value. Nutritional needs of the growing population.

Unit II

Food Fermentation Technology- Use of microorganisms to produce and preserve food products. Fermented foods - curd, cheese, bread, and vinegar have improved taste, shelf life, and nutritional value.

Unit III

Food Processing and Preservation- Food processing and preservation techniques help maintain food quality and prevent spoilage. Biotechnology contributes to the development of safer and longer-lasting food products through various preservation methods.

Unit IV

Nutritional Improvement of Foods Biotechnology is used to enhance the nutritional content of food through biofortification and genetic modification. Nutrient-enriched crops help reduce deficiencies and improve public health

Unit V

Functional foods and nutraceuticals - provide health benefits beyond basic nutrition, bioactive compounds that help prevent diseases and promote overall health and Well being.

REFERENCES

1. "Food Biotechnology "by Elsayed Abdel-Aaland And yKhatwa (2019)

2. "Introduction to Food Biotechnology" by Perry Johnson-Green(2016)

3. "Food Biotechnology "by Kalidas Shetty and Gopinadhan Paliyath (2005)

4. "Food Biotechnology, Second Edition "by Klaus Buchholz and Volker Kasche (2013)

5. "Nutritional Biochemistry and Metabolism: With Clinical Applications"by Maria Luz Fernandez and Jose M. Ordovas (2014)

6. "Biotechnology in Functional Foods and Nutraceuticals" edited by Debasis Bagchi and Francis, C. Lau (2010)

OUTCOMES

Understand food and nutrition biotechnology principles.

Explain microbial and enzymatic applications in food processing.

Apply biotechnology for food quality improvement.

Analyze nutritional and health benefits of foods.

Evaluate functional foods and nutraceuticals.

Understand food safety and modern biotechnological developments.

SECOND YEAR	CORE COURSE – VII	COURES CODE: 25CCCC7
SEMESTER - IV	OPERATING SYSTEM	CREDIT:5

COURSE OBJECTIVE:

- To recognize the concepts and principles of Windows operating system.
- To provide students with the basic knowledge and skills of memory in Windows
- To provide device and Process management and techniques in Windows environment.
- To able to do server monitoring of Windows
- To able to create group policy in Windows 2012.
- Install MS Windows 2012 environment.

UNIT	CONTENTS	HOURS
I	Windows-Hardware Basics, Operating System overview and Windows, Windows 7 Essential, Client OS-Windows 7 - Users and Groups-IP Configuration, Client OS-Windows 7 Tools and Utilities-Client OS Windows 7- Installation-Features-Disk Management-File Systems.	15
II	Server OS-Windows Server 2012 Overview-Server DNS-Zone Creation - DHCP - PRACTICAL-Advanced server storage Management-server ADS concepts and FSMO-Server OS Windows Server 2012 Roles and features-Server OS Windows Server 2012 File and Print Services.	15
III	Server OS monitoring and managing Windows Server 2012-Server OS Windows Server 2012 DNS and DHCP- Server OS Windows server 2012 Administrative Tools and ADS	15
IV	Server OS-Windows Server 2012-Storage and Backup Management-Client OS Windows 7Devices and Printers-Server OS Windows Server 2012 Installation.	15
V	Group Policy Management-Server Windows Server 2012- File and print services-Group Policy-Server Storage Management –Server Scenario-Server OS Windows Server 2012-DNS and DHCP -Server- ADS scenario.	15

COURSE OUTCOMES:

- Ability to work in Windows 7 operating system, its tools and utilities.
- Install and configure Windows 2012 server and client.
- Use administrative tools and backup management for Windows 2012 server.
- Understand device and Process management and techniques in Windows environment.
- Monitor server OS of Windows
- Create group policy in Windows 2012.

Text Books

1. William Stallings, *Operating Systems: Internals and Design Principles*, 9th Edition, Pearson Education, 2018.
2. Andrew S. Tanenbaum & Herbert Bos, *Modern Operating Systems*, 4th Edition, Pearson Education, 2015.
3. Greg Tomsho, *Guide to Operating Systems*, 5th Edition, Cengage Learning, 2017.

Reference Books

1. Thomas W. Shinder, *Windows Server 2012 Unleashed*, Sams Publishing, 2013.
2. Mark Minasi, *Mastering Windows Server 2012 R2*, Sybex Publications, 2014.
3. Ed Bott, Carl Siechert & Craig Stinson, *Windows 7 Inside Out*, Microsoft Press, 2011.
4. Abraham Silberschatz, Peter B. Galvin & Greg Gagne, *Operating System Concepts*, 10th Edition, Wiley India, 2018.

Web Sites

1. https://learn.microsoft.com/en-us/windows-server/
2. https://www.youtube.com/@NetworkChuck
3. https://www.guru99.com/operating-system-tutorial.html

SECOND YEAR	CORE COURSE – VI	COURES CODE: 25CCCP6
SEMESTER - IV	OPERATING SYSTEM LAB	CREDIT:2

COURSE OBJECTIVE:

- To recognize the concepts and principles of Windows operating system.
- To provide students with the basic knowledge and skills of memory in Windows
- To provide device and Process management and techniques in Windows environment.
- To able to do server monitoring of Windows
- To able to create group policy in Windows 2010.
- Install MS Windows 2010 environment.

EXERCISE	CONTENTS
1.	Windows 10 OS a. Windows 10 Installation b. Creating Users c. Configuring Disks d. Exploring File Permissions
2.	Windows Server OS a. Installation of Windows Server OS b. Installation of different Roles and Features
3.	Managing Windows Servers a. Create Users and Groups b. Explore Roles and Access c. Explore Network Configuration d. Explore Storage Management
4.	Monitor Windows Servers a. Exploring Task Manager b. Exploring Performance Monitor c. Exploring Event Log d. Exploring Alerts and Reports

5.	Backup & Recovering Windows Servers a. Backup Windows Server b. Restore Windows Server Backup

COURSE OUTCOMES:

- Ability to work in Windows operating system, its tools and utilities.
- Install and configure Windows 2010 server and client.
- Use administrative tools and backup management for Windows 2012 server.
- Understand device and Process management and techniques in Windows environment.
- Monitor server OS of Windows
- Create group policy in Windows 2010.

Software Essentials: OS – Windows/Linux

Web References

Practical:

1. https://www.youtube.com/watch?v=CraR01ya9ds
2. https://it.hessercan.com/wp-content/uploads/2017/09/laboratory_01-2017.pdf
3. https://docs.microsoft.com/en-us/windows/deployment/windows-10-poc

COURSE CODE	COURSE TITLE	YEAR	SEMESTER	CREDIT
	BIOTECHNOLOGY FOR HUMAN HEALTH	II	IV	2

Objectives:

- To develop new medicines, vaccines, and diagnostic methods for diseases
- To improve human health through genetic engineering and modern biotechnology techniques
- To prevent and treat infectious and genetic diseases effectively

Unit I

Industry Enzymes for textile industry, breweries, food supplements – single cell protein, vitamins, food processing - cheese, yoghurt making, biodegradable plastics, biofuels.

Unit II

Environment Applications of Biotechnology in environmental aspects: waste management, biodegradation of heavy metals, water cleaning, removing oil spills, air and soil pollution, bioremediation, biomining.

Unit III

Environment management - Environmental Impact Assessment - waste minimization techniques - environmental planning in urban development - natural resources and sustainable development - environmental ethics. tion, Increased milk production, artificial insemination, poultry and fisheries.

Unit IV

Applications in livestock improvement: transgenic animals, animal vaccine production .Transgenic crops for human

Unit V

Human Health and livestock Applications .Antibiotic production, Molecular diagnostics, vaccines and vaccine delivery, recombinant therapeutics – insulin, gene therapy, forensics.

Text Books/References

1. Bhasin, M.K. and Nath, S. 2002. Role of Forensic Science in the New Millennium, University of Delhi,

2. Crueger Wand Crueger, A. 2000. Biotechnology: A textbook of Industrial Microbiology. 2nd edition. Panima Publishing Co. New Delhi.

3. Eckert, W.G. and Wrightin, R.K. 1997. Introduction to Forensic Sciences, 2nd Edition, CRC Press, Boca Raton.

4. Hans-Joachim Jordening and Jesef Winter, 2005. Environmental Biotechnology Concepts and Applications.

5. James, S.H. and Nordby, J.J. 2005. Forensic Science: An Introduction to Scientific and Investigative Techniques, 2nd Edition, CRC Press, Boca Raton.

SECOND YEAR	ALLIED COURSE-I	CREDIT :
SEMESTER - III	ALLIED PHYSICS-I (MICROPROCESSOR & MICROCONTROLLER)	Subject code:

(II B.Sc., CSCS Students only**)**

Objective :

To Understand the Architecture and Memory Interfacing of microprocessor, Instruction set Interrupts ,Interfacing with simple I/O.

UNIT I: 8085 Microprocessor Architecture and Memory Interfacing: (12 Hours)

Introduction to microprocessor – 8085 microprocessor architecture – Pin out diagram-Timing diagram for op-code fetch, Memory read, Memory write, I/O read, I/O write operations.

UNIT II: 8085 Instruction set: (12 Hours)

Instruction classification – Instruction format –Addressing modes, Data transfer operation – Arithmetic and logical operations- Branching operations – Stack – Subroutines.

UNIT III: Interrupts: (12 Hours)

8085 interrupt-RST instruction – vectored and pending Interrupts

Interfacing with simple I/O: (12 Hours)

LED interfacing – switches-Memory mapped I/O-I/O mapped I/O

UNIT IV: Microcontroller (8051) Architecture: (12 Hours)

8051 hardware – Block diagram –oscillator and clock-PC and DPTR-A&B CPU - internal memory-internal RAM-stack and stack pointer -External memory-Connecting external memory.

UNIT V: Simple programs for 8085 (12 Hours)

8 bit addition and subtraction for 8085 - 8 bit multiplication for 8085 - 8 bit division for 8085 -16 bit addition and subtraction for 8085.

Total Hours- 60

TEXT BOOK:

1. The 8051 Microcontroller architecture programming and applications II edition – Kenneth J.Ayala- Penram international publishing, Rs 350.

2. The 8051 Microcontroller embedded systems II edition – Muhammed Ali Mazidi,JanishGillispieMazhidi-Person education publishing

REFERENCE :

The 8051 Microcontroller architecture programming and applications-Raj kamal

WEBSITES:

WWW.Atmel.com,WWW.microchip.com

Syllabus Prepared Staff Name: K.PREMKUMAR

SECOND YEAR	ALLIED COURSE-II	CREDIT :
SEMESTER - IV	ALLIED PHYSICS PRACTICAL -I (DIGITAL ELECTRONICS & MICROPROCESSOR)	Subject code:

(II B.Sc., CS CS Students only**)**

Objective:

It promotes the exhaustive requirements and expectations of the students to acquire practical knowledge for the theory given in their syllabus.

I. DIGITAL ELECTRONICS

1. Study of Logic gates using ICs (AND, OR, NOT, NAND, NOR, EX-OR).

2. Verification of De-Morgan's Theorem.

3. Universal building blocks-NAND

4. Universal building blocks- NOR.

5. Verification of Boolean expressions.

6. Half adder and full adder (using ICs).

7. Half subtractor and full subtractor (using ICs).

II. MICROPROCESSOR 8085

1.8-Bit addition & subtraction

2.8-Bit multiplication

3. 8-Bit division.

4. Sum of series.

5. Data transfer.

6. 16-Bit addition & subtraction.

Total Hours- 60

SECOND YEAR	NON-MAJOR ELECTIVE- II	COURSE CODE: 25ITNM2A
SEMESTER - IV	WORKING PRINCIPLES OF INTERNET	CREDIT: 2

COURSE OBJECTIVES:

- To teach the basics of the World Wide Web
- To understand the fundamentals of the Internet and the usage
- To know the components of Multimedia on the internet

UNITS	CONTENTS	HOURS
I	What is Internet? The Internet's underlying Architecture	5
II	Connecting to the Internet – Communicating on the Internet	5
III	How the World Wide Web works. Common Internet tools	5
IV	Multimedia on the Internet – Intranet and shopping on the Internet	5
V	Safeguarding the Internet	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the evolution of the Internet.	K2
CO2	Know the basic knowledge of the web	K4
CO3	Comprehend the protocols and standards used throughout the Internet.	K2
CO4	Discuss a variety of Internet and WWW applications and related technologies.	K4
CO5	Evaluate the opportunities and threats created by interconnecting computers via the Internet.	K5

REFERENCES:

1. Preston Gralla, "How the Internet Works", Pearson Education, Eighth Edition, 2006.
2. C.Xavier, Fundamentals of Internet and Emerging Technologies, New Age International Private Limited; First Edition ,2021
3. Alexis Leon, Internet for Everyone, S. Chand (G/L) & Company Ltd; Second Edition 2012.
4. Andrea C. Nakaya,"Internet and Social Media Addiction", Reference Point Press,2015.
5. Richard Fox, Wei Hao,"Internet Infrastructure: Networking, Web Services, and Cloud Computing", CRC Press, 2017.

SECOND YEAR	NON MAJOR ELECTIVE II	COURSE CODE: 25ITNMS2B
SEMESTER - IV	WEB DESIGNING	CREDIT: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.

UNIT	CONTENTS	HOURS
I	Introduction to Web related terms: Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	5
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator	5
III	Web Browser: Web Index – Web Search Engine–Meta Search Sites–Directory and Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail	5
IV	Introduction to HTML: HTML Code for Web Page–HTML Tags-Web Page Basics–Display a Web Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule	5
V	Introduction to HTML Table: Cell Padding – Cell Spacing – Table Borders, HTML–Advantages and Disadvantages.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Explain the fundamentals of Internet, WWW, web architecture, and networking concepts.	K1
CO2	Identify and describe various Internet protocols and their roles in web communication.	K3
CO3	Use web browsers, search engines, and email systems effectively for information retrieval and communication.	K3
CO4	Design basic web pages using HTML elements such as text formatting, lists, images, and links.	K4
CO5	Create and manage HTML tables and apply formatting techniques for structured data representation.	K5

REFERENCES:

1. Alexis Leon & Mathews Leon, "Internet for Everyone", Leon Tech World, Chennai.2, 2009
2. John Zabour,JeffFoust & David Kerven,"HTML 4 HOW-TO",2005
3. Eric Kramer,"HTML". Kamalesh N.Agarwala, AmitLal & Deeksha Agarwala,"Business of the net, 2008.
4. Responsive Web Design with HTML5 and CSS (5th Edition) – *Ben Frain, Packt Publishing, 2025*